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Q2.5) The video explains how to plot and the data needed for Levenspiel plots for CSTR, PFR, and batch reactor. As well as the equations and methods/ideas used.

The speaker did a great job emphasizing the equations and methods used. He could be clearer showing the graph.

P2.2)

1) 10, 50, 80% conversion $n = 100 \text{ mol}$ of A through 400 dm^3 reactor

$$C_{A,0} = \frac{N_{A,0}}{V} = \frac{100}{.4} = 250 \frac{\text{mol}}{\text{m}^3}$$

$$\int_0^{.1} \frac{dX}{-r_A} = \frac{.1}{2} \left(\frac{1}{.45} + \frac{1}{.37} \right) = .246 \frac{\text{m}^3 \text{s}}{\text{mol}} \quad t_{.1} = (250)(.246) = 61.5 \text{s}$$

$$\int_0^{.5} \frac{dX}{-r_A} = 2.01 \frac{\text{m}^3 \text{s}}{\text{mol}} \Rightarrow t_{.5} = (250)(2.01) = 503 \text{s}$$

$$\int_0^{.8} \frac{dX}{-r_A} = 5.01 \frac{\text{m}^3 \text{s}}{\text{mol}} \Rightarrow t_{.8} = (250)(5.01) = 1250 \text{s}$$

2) $\underline{X = .8}$
CSTR: 6.4 m^3
PFR: 2.165 m^3

$$F_{A0} \text{ cut in half} \Rightarrow V_{\text{new}} = V_{\text{old}} \left(\frac{.2}{.4} \right)$$

$$\text{CSTR: } V = 3.2 \text{ m}^3$$

$$\text{PFR: } V = 1.0825 \text{ m}^3$$

$$\text{doubled} \Rightarrow V_{\text{new}} = V_{\text{old}} \left(\frac{.8}{.4} \right)$$

$$\text{CSTR: } 12.8 \text{ m}^3 = V$$

$$\text{PFR: } 4.33 \text{ m}^3 = V$$

$$\text{Conversion: } V = \frac{F_{A0} X}{-r_{A, \text{exit}}}$$

linear
interpolation

$$X \approx .73$$

$$X_{\text{PFR}} = 0.90$$

3) $V = 2.5 \text{ m}^3$ CSTR ; $F_{A0} = 0.4 \frac{\text{mol}}{\text{s}}$

X	.6	.7
$-r_A$	.113	.079

$$2.5 = \frac{.4X}{-r_A(X)}$$

$\downarrow \quad \downarrow$
 $V = 2.12 \text{ m}^3 \quad V = 3.54 \text{ m}^3$

$$X \approx .6 + \frac{2.5 - 2.12}{3.54 - 2.12} (.1) = 0.63$$

63% conversion

4) $V = 1.0 \text{ m}^3$ PFR w/ $V = 2.5 \text{ m}^3$ CSTR

$$V = \int_{x_1}^{x_2} \frac{F_{A0}}{-r_A(X)} dX$$

X	$\frac{F_{A0}}{-r_A(X)} dX$
.6	3.54
.7	5.06
.8	8.00

$h = 6 \text{ m}^3$
 $w = .17 \text{ m} \Rightarrow A = 1.0 \text{ m}^3$

$$X_2 = 0.79$$

79% conversion

5)

$F_{A0} = .4 \frac{\text{mol}}{\text{s}}$	$V_1 = .82 \text{ m}^3$	$V_i = \frac{F_{A0,i} X_i}{(-r_A) X_i}$
$F_{A0,i} = .2 \frac{\text{mol}}{\text{s}}$	$V_2 = 3.20 \text{ m}^3$	

$$X_1 = .4 + \frac{1.64 - .82}{2.12 - .82} (.2) = 0.53$$

$$V_2(.2) = 3.2 \text{ m}^3 \Rightarrow 2(3.20) = 6.40 \text{ m}^3 \text{ so } X_2 = .80$$

$$\text{total conversion} = \frac{0.53 + .80}{2} = 0.663$$

66.3% conversion

6) Worst order: PFR to CSTR to CSTR

$$V_1 = \frac{1}{3}(6.54) = 0.218 \text{ m}^3 \quad V_2 = 2.795(.3) = 0.8385 \text{ m}^3$$

$$V_3 = 8.00(.3) = 2.40 \text{ m}^3$$

$$V_t = 3.460 \text{ m}^3$$

$$X_1 = .1 + .00139 = .1014$$

$$X_2 = 0.180 \quad X_3 = .246$$

$$7) V = 3 \text{ m}^3 \quad v_0 = \frac{3}{2} = 1.5 \frac{\text{m}^3}{\text{s}} = 90 \frac{\text{m}^3}{\text{min}}$$

2.10) 1) PBR to CSTR

$$2) W = 23.2 \text{ kg of catalyst (given)}$$

$$3) W = \left(\frac{F_{A0}}{-r_A} \right) X = (12 \text{ kg}) (.4) = 4.8 \text{ kg}$$

$$4) W = \int_0^{.8} \frac{F_{A0}}{-r_A} dX \quad \text{graph gives } W_{\text{PBR}, .8} \approx 16 \text{ kg}$$

$$5) W \approx 3.5 \text{ kg}$$

Q3.2)

Chapter 3: Rate Laws

Self Test: 1 Clicker questions - Multiple Choice Questions

Question:

3) The rate of the formation of NO_2 in the reaction

$$2\text{NO} + \text{O}_2 \rightleftharpoons 2\text{NO}_2$$

is $r_{\text{NO}_2} = 4 \text{ moles/m}^3/\text{s}$

The rate of disappearance of O_2 , $-r_{\text{O}_2}$

A) $-r_{\text{O}_2} = -4 \text{ moles/m}^3/\text{s}$

B) $-r_{\text{O}_2} = +8 \text{ moles/m}^3/\text{s}$

☒ C) $-r_{\text{O}_2} = +2 \text{ moles/m}^3/\text{s}$

D) $-r_{\text{O}_2} = -2 \text{ moles/m}^3/\text{s}$

MacBook Pro

Could be included as a basic question, however not complex enough to stand alone.

P3.12)

$$1) k = \left[\frac{\text{dm}^3}{\text{mol} \cdot \text{h}} \right] \quad -r_A = k C_A C_B \quad A + B \rightarrow C$$

$$2) k = \left[\frac{\text{mol}}{\text{kg cat} \cdot \text{h} \cdot \text{atm}^2} \right] \quad -r'_A = k P_A P_B \quad \text{for } A + B \rightarrow C$$

$$3) k = \left[\frac{1}{\text{h}} \right] \quad -r_A = k C_A \quad \left[\frac{1}{\text{h}} \right] \left[\frac{\text{mol}}{\text{dm}^3} \right] = \left[\frac{\text{mol}}{\text{dm}^3 \cdot \text{h}} \right] \quad \text{for } A \rightarrow B$$

$$4) k = \left[\frac{\text{mol}}{\text{dm}^3 \cdot \text{h}} \right] \quad -r_A = k \quad \text{zero order / nonelementary}$$